

RiC Weekly Note #2

Lecture Content

- Introduction to motion planning
- Bug Algorithms

Literature

- Chapter 1 and 2 in “Principles of Robot Motion”

Exercises

1. What are the different properties of a planning algorithm?
2. What are the tradeoffs between Bug 1 and Bug 2? You could consider aspects such as
 - Best case / worst case performance, e.g., path length
 - Completeness
3. For the Bug 1 algorithm: argue whether the robot will ever visit an obstacle that is not within the circle radius $d(q_{\text{start}}, q_{\text{goal}})$ with center in q_{goal} .
4. What assumptions are used in the bug algorithm when considering localization? Are these reasonable (realistic)?

Programming Exercise 2

See separate file on ItsLearning